

Database Concepts -- Guided Notes ANSWER KEY

Unit tp-5.4 | CompTIA Network+ N10-009 Obj. FC0-U71 5.4 | N10-008 FC0-U71 5.4

For instructor use / distribute after students complete the notes.

Question 1

A relational database stores data in _____ with rows and columns. The relationship between tables is defined using _____ keys that reference _____ keys in other tables. This structure prevents data _____ and ensures that related data stays consistent.

Answer: Tables; foreign; primary; duplication (redundancy).

Question 2

ACID properties ensure reliable database transactions. A = _____ (all steps complete or none do), C = _____ (data remains valid), I = _____ (transactions do not interfere with each other), D = _____ (once committed, data persists even after a crash).

Answer: Atomicity; consistency; isolation; durability.

Question 3

A database _____ is a saved SQL query treated as a virtual table. It does not store data itself -- it runs the query dynamically. A database _____ is code stored in the database and executed on demand, allowing complex logic to run _____ to the data without round trips to the application server.

Answer: View; stored procedure; close (server-side).

Question 4

An index in a database works like a book index -- it allows the DBMS to find records _____ without scanning every row. Creating an index on a frequently queried column speeds up _____ operations but slightly slows _____ and _____ operations because the index must be maintained when data changes.

Answer: Faster; SELECT (read); INSERT; UPDATE/DELETE (write).

Question 5

Database _____ is the process of copying data to another location to protect against data loss. _____ backups copy all data each time. _____ backups copy only data changed since the last full backup. _____ backups copy only data changed since the last backup of any type. The _____ (Recovery Point Objective) defines the maximum acceptable data loss window.

Answer: Backup; full; differential; incremental; RPO.

Question 6

A database _____ creates linked copies of the database on multiple servers for redundancy. A _____ replication model has one server that accepts writes and pushes changes to read-only _____. This improves _____ availability and _____ performance by distributing read queries.

Answer: Replication; primary-replica (master-slave); replicas; high; read.

Question 7

NoSQL databases trade strict _____ consistency for horizontal _____ (spreading data across many servers). The CAP theorem states that a distributed system can guarantee only _____ of these three: Consistency, Availability, and Partition Tolerance. Most NoSQL systems choose _____ and Partition Tolerance (AP) over strong consistency.

Answer: ACID; scalability (scaling); two (2); availability.

Question 8 -- Real World Application

REAL WORLD: An e-commerce website uses a single database server with no backup and no redundancy. One morning the server hard drive fails. The website is offline and all customer order history is gone. Describe three database infrastructure practices the company should implement to prevent this scenario from recurring.

Answer: Practice 1 -- Automated scheduled backups: configure daily full backups and hourly incremental backups stored on a separate server or cloud storage (e.g., AWS S3). Test restores quarterly to verify backups are functional. With hourly incrementals, maximum data loss is 1 hour of orders. Practice 2 -- Primary-replica replication: configure a replica (read-only copy) on a second server that stays synchronized with the primary database in near real-time. If the primary drive fails, failover to the replica -- it has all committed transactions. Downtime is minutes rather than hours. Practice 3 -- RAID on the database server: configure RAID 1 (mirroring) or RAID 5/6 on the server so that even if one physical drive fails, the database continues operating on the remaining drives with no data loss. This protects against drive failure without any downtime.